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A review of cybersecurity in smart cities and intelligent transport systems

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Abstract

The increasing uptake of digital technologies in smart cities and intelligent transport systems (ITS), has increased associated cybersecurity concerns, with cyber threats on critical infrastructure, public safety, and data privacy. This bibliometric analysis studies cybersecurity research in smart cities and ITS over the last ten years (2015–2025) through 170 relevant documents retrieved from Scopus database. It focusses on research themes that include cybersecurity risk management, AI-driven threat detection, IoT security in urban environments, policy and governance frameworks, and resilience strategies against cyber threats. The use of blockchain technology in a decentralized way for maintaining data integrity, secure communication, and transparent transaction validation among smart city networks has particularly become a major concern. A pattern of trend of publication suggests an increase in research on cybersecurity issues after 2020 with greater interdisciplinary collaborations, data encryption, real-time anomaly detection, and regulatory compliance. Research studies continue to highlight persistent challenges in IoT device vulnerabilities and the lack of a standardized security protocol for securing large-scale urban networks. The results also tend towards AI-enhanced security models, blockchain-based authentication mechanisms, and policy-driven cybersecurity frameworks. Future research directions include quantum-resistant cryptographic solutions, AI-driven predictive security models, and cross-sector regulatory initiatives that can enhance cybersecurity resilience in smart urban infrastructures.

Article Highlights

- Proposes a unified model linking AI, blockchain, and IoT security interactions in smart cities.
- Outlines key themes on AI security, blockchain authentication, and IoT-enabled intelligent mobility systems.
- Identifies security gaps and highlights future directions such as quantum-safe cryptography and AI-based cyber defence.

Keywords Smart cities, Intelligent transport systems, Cybersecurity, AI in cybersecurity, Critical infrastructure protection, Cyber threats



Graphical Abstract

Literature search	Methodology	Findings Proposed Model	Findings Key themes	Findings Key clusters	Future research directions
- Articles of 170 SCOPUS articles from 2015-2025 - Analysis Includes: - Identification of key themes. - Identification of key trends	- Bibliometric analysis. - PRISMA. - VOS Viewer. - Biblioshiny - Thematic analysis	Proposed taxonomy/ model linking AI-Blockchain-IoT security interactions in smart cities.	- Cybersecurity" and "smart city - Internet of things - Network security - Cyber-attacks" and "intrusion detection - Risk assessment - Cryptography - Blockchain - Data Privacy	3 clusters: - Privacy and Ethical Issues - Artificial Intelligence and Cybersecurity: - Blockchain in case of Secure Smart Cities - Privacy and Ethical Issues	- Quantum-resistant cryptographic solutions, - AI-driven predictive security models, and - Cross-sector regulatory initiatives

1 Introduction

The developing modernisation of urban environments has introduced smart cities and intelligent transport systems (ITS) that accomplish higher efficiencies, sustainability, and quality of urban living. Smart cities exploit emerging technologies such as the Internet of Things (IoT) and Artificial Intelligence (AI) to facilitate the management of urban services, allocate resources more effectively, and improve mobility. Intelligent Transportation Systems (ITSs) have emerged as a revolutionary force in modern transportation's fast-expanding environment, fundamentally changing how we transport people and things. They combine cutting-edge technology, data analytics, and communication systems that aim to improve transportation networks' effectiveness, security, and environmental friendliness [1]. Robust cybersecurity frameworks will thus play a vital role in securing and protecting smart urban ecosystems from these threats as urban digitalisation grows exponentially.

Cybersecurity research related to smart cities and ITS has gained significant attention over the last decade. This paper is a bibliometric analysis of 170 research works indexed in Scopus from 2015 to 2025, centred on key trends in urban cybersecurity. Relevant literature studies have extensively focused on securing IoT-enabled smart grids, real-time traffic monitoring systems, and public surveillance networks [2]. Interconnection in such systems constitutes multiple attack vectors and hence is subjected to network intrusion attacks, unauthorized access, and cyber-physical attacks [3]. As urban infrastructures become increasingly data-driven, protecting critical services from cyber threats is paramount to maintaining operational stability and public safety.

The challenge of securing smart cities arises from the multiplicity of devices and systems existing in an urban environment. The IoT-enabled sensors, connected devices, and cloud-based services function over multiple platforms, none of which follows standard security protocols. This fragmentation increases vulnerabilities, making it challenging to impose uniform cybersecurity measures across various sectors of an urban environment [4]. IOT devices have limited memory which also limit the ability to implement strong encryption mechanisms in smart city applications. To overcome these challenges and improve cyber resilience in smart urban environments, researchers have suggested AI-driven anomaly detection, blockchain-based authentication.

Another critical component of smart cities is ITS, which is also vulnerable to cybersecurity threats. ITS technologies, such as connected and automated vehicles (CAVs), smart traffic lights, and intelligent public transportation systems, rely on real-time data exchange to optimize mobility [5]. However, the interconnection between wireless communication networks and cloud-based platforms risks exposure to cyberattacks, thereby exposing an interference in traffic management and putting the lives of commuters at risk [6]. The vulnerability associated with vehicle to everything (V2X) communication systems, raises further questions regarding data integrity, unauthorized access, and remote hijacking capabilities of a vehicle [7]. Studies have emphasized the need for secure authentication protocols, AI-driven intrusion detection, and quantum-resistant cryptographic solutions to protect ITS from emerging cyber threats.

For example, the European Union General Data Protection Regulation imposes strict data privacy measures to ensure that citizens' personal information is protected from cyber threats. The National Institute of Standards and Technology in the United States also offers cybersecurity frameworks that can help municipalities establish security best practices for smart infrastructure [1]. Despite all these efforts, there is still no universal standard for cybersecurity that is specifically designed for smart cities, which means that the implementation of policies across different regions may not be consistent.

Bibliometric analysis of 170 research papers from the last ten years shows an increase in academic interest in urban cybersecurity. The number of publications on the theme of smart city security increased post-2020, revealing an urgency of concern in tackling cybersecurity threats within digital urban ecosystems. Studies also report a recent rise in inter-disciplinary research collaboration between cybersecurity experts, urban planners, and policymakers to establish all-rounded security strategies [2]. In addition, new research trends of AI-powered cybersecurity solutions, blockchain-based identity management, and zero-trust architectures in secure smart urban environments are gaining momentum.

Indeed, a huge research gaps exist regarding one of the main focal challenges, i.e., legacy infrastructure in cities going through digitalization. Most of the urban systems in this regard are using old security and are exposed to cyber-attacks. This is in addition to the difficulty of incorporating various security frameworks across interconnected urban services. Policymakers and city administrators face such challenges [3].

The research questions in this study will address as follows:

RQ 1. What are the main cybersecurity threats to smart cities and intelligent transport systems?

RQ 2. How have the contributions of cybersecurity research undergone restructuring regarding publication trends, citations, and international collaborations?

RQ 3. What are the trends of emerging cybersecurity frameworks, including AI-level security models and blockchain implementations?

RQ 4. What gaps and challenges face urban digital infrastructure from a security standpoint?

A bibliometric analysis will, therefore, provide insightful arguments on the evolution of cybersecurity strategies in smart urban environments. The identified research clusters help to contribute the existing body of knowledge and through the identification of influential authors and policy developments that continue to explain cybersecurity advancement in smart cities and ITS. This is especially useful to help researchers, policymakers,

and urban planners develop resilient and secure urban digital ecosystems. By conducting a bibliometric analysis, this research provides valuable insights into the evolution of cybersecurity strategies in smart urban environments.

The study begins by giving an overview of cybersecurity in smart cities and ITS in the introduction in the first section, followed by an in-depth literature review in the second section. A structured research methodology has been defined and deployed for analysis in the third section. This is followed by results; discussions; and conclusions, implications, and limitations for the study.

2 Literature review

Development of intelligent transport systems and AI has brought the world to unprecedented dependence on digital technologies. These developments increase the efficiency of cities but raise a significantly large threat level in cybersecurity. Thus, research has dealt with protecting critical urban infrastructure from cyber threats, having proper data privacy, and providing regulatory regimes for newly arising issues [5, 8]. By analyzing recent developments and theoretical perspectives, this literature review provides a detailed report understanding of the cybersecurity landscape in smart cities and ITS.

2.1 Cybersecurity in smart cities: theoretical perspectives

Cybersecurity literature takes into consideration different perspectives on securing smart city infrastructure, intelligent transport systems (ITS) as well as urban digital ecosystems [9]. The increase in the integration of IoT, AI, and cyber-physical systems in urban environments has escalated levels of concern regarding cyber threats, data breaches, and risks to privacy [10]. In exploring urban cybersecurity, research utilized a variety of frameworks that range from risk management strategies and encryption frameworks to AI-driven anomaly detection and broader regulatory policies.

The Cybersecurity Threat Model suggests that cyber risks in smart cities originate from interconnected digital systems that can be exploited by malicious actors. This model highlights the need for a multi-layered security approach, incorporating initiative-taking threat detection, real-time monitoring, and incident response mechanisms [9].

The Privacy Preservation Theory is also relevant to smart cities, which would focus on measures of data protection, encryption, and privacy-enhancing technologies to prevent unauthorized access to sensitive urban data. Further, the adoption of Zero-Trust Architecture reinforces security, with no entity in the network considered inherently trusted, thus reducing internal and external cyber threats.

Hence, the following literature review categorizes cybersecurity challenges into four subsections:

2.2 Cybersecurity threats in smart cities

The increasing reliance on the digital infrastructure of smart cities has heightened the dangers of cyber threats. Studies have shown that urban services are required to be protected by some integral cybersecurity frameworks [8]. Research studies indicate vulnerabilities in CPS and IoT-enabled infrastructure with AI-powered security mechanisms acting as a potential counter measure. These vulnerabilities arise from the increased connectivity of smart city systems, which leaves them vulnerable to cyber-attacks [10].

2.3 Privacy and data security in smart cities

Emergence of IoT in urban settings has opened box of excessive data gathering on the one hand while raising serious concerns about the fact of infringement on privacy and unauthorized access to it. Smart city applications rely on vast amounts of real-time data generated by a number of setups, including surveillance cameras, traffic monitoring, and the optimization of public services. These serve as a basis for considerable privacy problems from the leaked data available for abuse. Some authors suggested that the use of blockchain security frameworks with zero-trust architectures could address some of these concerns quite effectively [9].

Blockchain technologies protect the integrity of data through decentralization of information storage, which means that such data is out of reach from centralized targeting attacks. AI-powered encryption models provide another layer of data protection against cyber threats in smart cities. Countries around the world have enacted data protection laws to ensure that smart city data is collected, stored, and used responsibly—for example, the General Data Protection Regulation (GDPR). Looking ahead, this shift toward compulsory protection of privacy will need to be nurtured from now onward so that innovation does not crook privacy's back. Public awareness and regulatory enforcement are another important dimension of data security in smart cities. Most citizens are not aware of how their data is collected and processed in smart environments.

2.4 Cybersecurity in ITS

Research in intelligent transport security emphasizes the need for encryption and intrusion detection to secure vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V) communications [10]. Transportation networks are increasingly reliant on digital technologies, making them vulnerable to cyber threats. Cyber-attacks targeting transportation infrastructure have been rising, advocating for AI-powered anomaly detection systems [9].

Autonomous and connected vehicles are among the most susceptible elements of ITS. Cybercriminals can exploit vulnerabilities in communication protocols to hijack vehicles, manipulate traffic signals, or disrupt transit systems. These threats may be countered with multi-layered cybersecurity approaches. Encryption, blockchain security, and machine learning-based threat detection are some options that researchers propose. Transport risks regarding cyber threats have also been looked at and reinforced further on the need for seamless system upgrades.

Another area of concern is that of public transport networks' security. Many cities have integrated digital payment systems and smart ticketing into their public transport offers. While easy for users, it introduces a way through which an attack can occur. Cybercriminals use weak encryption protocol environments to intercept financial transactions or steal a customer's sensitive information [5].

2.5 Future trends in cybersecurity for smart cities

Advances in 6G network, edge computing, and quantum cybersecurity bring forth numerous challenges and opportunities. This will be combined with systems in place to counter emerging threats, which would propose AI-driven automated threat response solutions [8]. As 6G networks develop, smart city infrastructures are expected to merge

further; however, this very enlargement is going to mass the attack surface thus intensifying security enforcement requirements.

Quantum computing poses yet another significant challenge to cybersecurity. While offering the potential for a game-changing evolution in encryption practice, quantum phenomenology enhances the risks posed to the older frameworks of cryptography. Being already damaged, present encryption techniques, for example RSA standard and ECC method, may not withstand quantum decryption. Researchers hold one view that the answer to the future crux of cyber security will be the advent of Post Quantum Cryptography (PQC).

Regulatory measures and compliance policies, GDPR and CCPA policies tremendously mold the landscape of cybersecurity. The collaboration of the government and industry stakeholders in establishing a standard security protocol for smart city applications is inevitable. Apart from the above, it must also be ensured that what is done is not contradicted by the ethics of data utilization and AI-driven decision-making processes in urban contexts.

Smart cities can protect the digital infrastructure from these attacks and thus ensure sustainability in urban development by expanding research and developing secure creative solutions. Future development in cybersecurity will guide us toward the creation of secure, resilient, and urban environments.

3 Research methodology

A structured and systematic bibliometric analysis is used in this study to analyze cybersecurity in smart cities and ITS. Systematic bibliometric tools facilitate the extraction of data, performance evaluation, and the creation of citation and co-citation networks for the investigation of the evolution of research in the field [11]. Bibliometric data were obtained from the Scopus database via the search string: (“cyber security” OR “transportation” OR “smart city”), which is composed of the publications with titles, abstracts or keywords relating to security of computer systems, transportation networks and urban research. The dataset was extracted on 10 February 2025, thus guaranteeing the use of a consistent temporal cut-off throughout the analysis. To further purify the dataset, certain criteria were set to include only peer-reviewed English publications from 2015 to 2025 and studies dealing with devices, systems, and technologies in smart cities which are the direct topics of urban innovation and intelligent transport field of cybersecurity. Papers not corresponding to the thematic focus or not fulfilling the language and time-frame criteria were rejected. A search reproducibility statement is upheld by the recorded query string and the date of extraction, which permits the search to be duplicated on the same Scopus settings. There were no extra duplicate filtering methods used as the first Scopus export did not include any repeated entries.

The research methodology employed citation analysis and content analysis, which are quite common in cybersecurity research. Citation analysis maps the relationship between cited and citing articles, allowing the identification of core literature, leading journals, and research clusters [12]. When anticipating smart city privacy and security challenges, it is important to realize that many of the same challenges exist today, albeit not as frequently as when these technologies become fully interconnected. Current deliberations and attempts to solve security and privacy challenges will lead to understanding how smart cities can be securely constructed [13].

To conduct this bibliometric study, data has been sourced from Scopus, one of the most widely recognized academic databases for bibliometric research [14]. Bibliometric software tools such as Biblioshiny and VOS viewer are employed for network analysis, co-word analysis, and collaboration mapping to visualize the intellectual structure of cybersecurity research. The PRISMA guideline was used for reporting the review studies (see Fig. 1).

As a result, the implications of new smart city security systems for crime reduction, security, and urban governance are rarely discussed. This systematic review attempts to address this gap by exploring the last ten years' worth of literature on new security technologies that can be considered to fall under the smart city concept.

3.1 Database selection

The Scopus database has been selected as the prime source for literature searching for its broad coverage, trustworthiness, and exhaustive citation profiling. Scopus is universally accepted as the most credible database in conducting bibliometric research studies in cybersecurity, smart cities, and intelligent transport [2]. This database provides access to high-impact peer-reviewed research articles, thus laying a solid background for this current study.

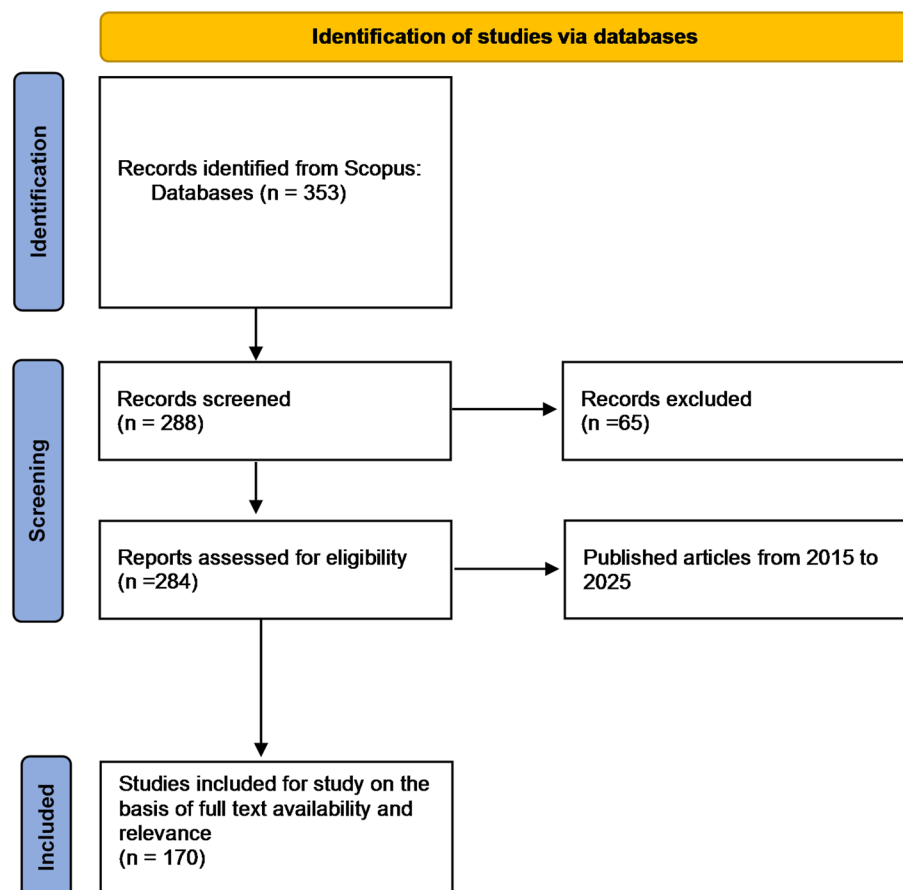


Fig. 1 PRISMA flow chart showing the inclusion and exclusion criteria. (Source: authors own work)

3.2 Search strategy

Considering collection of relevant literature and then extraction of the final dataset demanded a structured, multi-step approach. For refining the search strategy, thorough examinations of previous bibliometric studies in cybersecurity and smart cities were carried out [15]. From there, a shortlist of appropriate keywords emerged that would be put together to create the search string in Scopus. The searches were conducted by deploying diverse combinations of keywords, such as: “cyber security” OR “transportation” OR “smart city. The initial search yielded several hundred articles. However, not all articles were related to the study area. To ensure relevance, 170 research articles were carefully selected for final analysis based on pre-defined criteria.

3.3 Search refinements

To guarantee that only the most pertinent research articles were selected, the subsequent inclusion and exclusion criteria were utilized:

- (a) Only articles that have been published in peer-reviewed journals were considered.
- (b) Timeframe for analysis: Research articles published from 2015 to 2025 were incorporated to ensure up-to-date and pertinent insights in the fields of cybersecurity and smart cities.
- (c) Subject areas: The study focused on articles covering major subject areas such as computer science, engineering, cybersecurity, artificial intelligence, IoT, smart infrastructure, and data privacy to maintain a multidisciplinary approach.
- (d) Document type and language: Only research articles published in English were considered. Conference proceedings, preprints, working papers, and magazine articles were excluded.

After applying these refinements, the final dataset of 170 articles was saved in three separate formats:

CSV file: Used for analysis in Scopus and VOSviewer for citation, co-citation, and network analysis.

BibTeX file: Used for Biblioshiny to perform detailed bibliometric mapping and collaboration analysis.

Excel spreadsheet: Used for organizing and categorizing articles for key thematic clustering.

The bibliometric analysis was done using the Bibliometrix R package Biblioshiny (version 5.0), which allowed to the performance analysis, mapping of keyword co-occurrence and trends in publication over time. For the visualization of networks, such as co-authorship, co-occurrence, and country collaboration maps, the use of VOSviewer version 1.6.20 was made, which was chosen for its ability to produce high-quality graphical representations of the network.

This methodological approach ensures a structured and reproducible review of the literature, facilitating the identification of key research themes, influential authors, and potential research gaps in cybersecurity for smart cities and intelligent transport systems.

4 Analysis and results

The last 170 research papers were analyzed via three different methods: systematic bibliometric analysis, network visualization, and graphical analysis for keyword co-occurrence, country co-authorship, and research topics. Below are the results from each of these analyses.

4.1 Bibliometric analysis

The bibliometric review of cybersecurity for smart cities and intelligent transport uncovered 170 research articles, with overall citations of (3191), average citation per paper (18.63) and h-index distribution (453), which sought to address the research question RQ 2 and RQ 4. The articles were published in different academic journals covering a range of disciplines such as cybersecurity, IoT, artificial intelligence, smart infrastructure, and data privacy. Additionally, the research assessed research output from varied viewpoints such as chronological trends, author contributions, journal impact, country performance, and institutional collaboration.

4.2 Conceptual framework

The three research pillars AI-driven security, blockchain-enabled trust, and IoT infrastructure align directly with the layered interactions depicted in Fig. 2 depict the taxonomy for AI–Blockchain–IoT security interactions in smart cities. The emergence of citation and keyword clusters also points to a growing convergence of AI, blockchain, and IoT technologies which confirms a policy roadmap that supports the integration of multi-technology security architectures in smart cities as a priority area. This study, while acknowledging the prior bibliometric studies that focused on individual components like IoT security or smart city governance, maps the interdependencies across AI, blockchain, and intelligent transport systems, thus providing a comprehensive security framework that is in line with current research trends and technological synergies.

4.3 Chronological trends in research publications

Figure 3 displays the chronological distribution of 170 articles published between 2015 and 2025 which sought to address the research question RQ 1 and RQ 2. The trend of publications displays robust growth from 2021, with the maximum number of articles (44) being published in 2024.

Figure illustrates a 150% surge in publications from 2021 to 2024. This exponential increase reflects heightened industry focus on securing IoT infrastructure against post-pandemic cyber threats.

Additionally, the steep fall in 2025 (with just two articles reported) can be attributed to the partially available dataset for the year or an interim realignment of research agendas. The rising publication rate in earlier years attests to the increasing relevance of cybersecurity frameworks to counter smart city technology vulnerabilities.

4.4 Author impact and performance

In-depth analysis of the leading author contributors to smart city and intelligent transport cybersecurity research indicates a keen academic interest in cyber resilience, AI-based security, mitigation of IoT vulnerabilities, and blockchain-authentication protocols. Table 2 presents the leading ten authors with the largest number of research

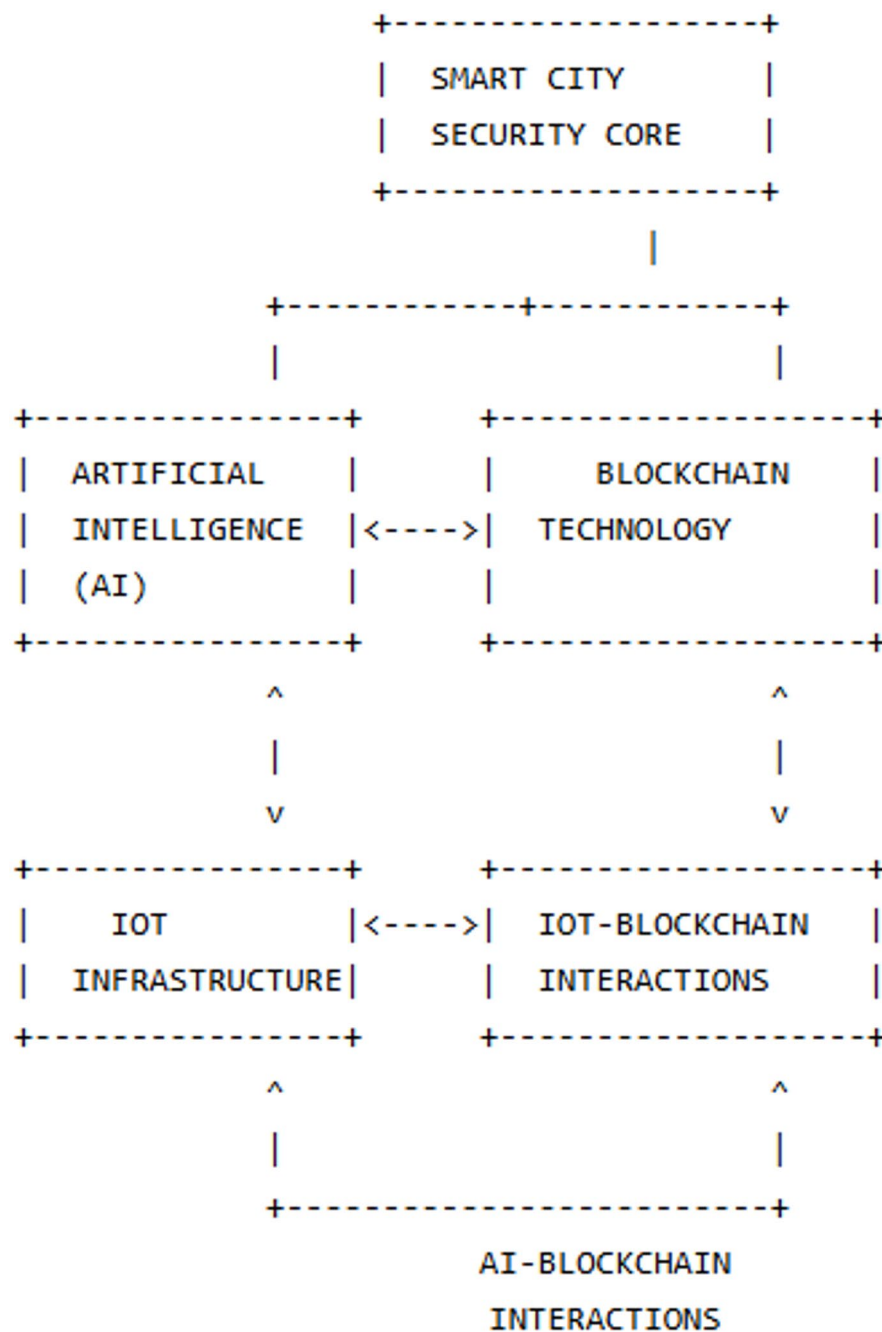


Fig. 2 Taxonomy of AI-blockchain-IoT security interactions in smart cities. (Source: authors own work)

papers published on this topic which sought to address the research question RQ 1 and RQ 3.

As can be seen from Table 1, the most active researchers, such as Ahmad F, Andrade RO, Kumar R, Nguyen HN, Shaikh SA, Taylor SJ, and Tello-Oquendo L, each have three articles published, with them being among the prominent contributors in the discipline. Their works have addressed:

Ahmad F & Kumar R: Mitigating smart infrastructure's cybersecurity vulnerabilities through a focus on cyber risk modeling, AI-driven attack detection, and threat intelligence exchange. Andrade RO & Nguyen HN: Examining machine learning use cases

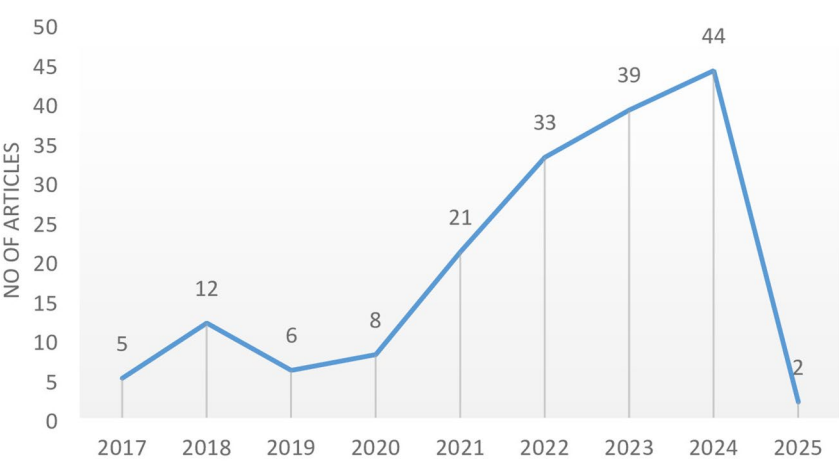


Fig. 3 Trends in research publication.

Table 1 Top 10 authors by number of publications. Source: the authors (using Scopus database)

Authors	Articles
AHMAD F	3
ANDRADE RO	3
KUMAR R	3
NGUYEN HN	3
SHAIKH SA	3
TAYLOR SJ	3
TELLO-OQUENDO L	3
ALHASHMI SM	2
AMRO A	2
ELLOUMI M	2

in cybersecurity, specifically anomaly detection methods, behavioral analytics, and predictive threat modeling. Shaikh SA & Taylor SJ: Emphasizing privacy-preserving architectures for intelligent transport, secure data transfer in IoT-enabled smart cities, and regulatory implications of AI-enabled cybersecurity. Tello-Oquendo L: Investigating blockchain-enabled security solutions for smart mobility and digital identity verification.

Moreover, Alhashmi SM, Amro A, and Elloumi M, who have each contributed two research papers, have also made significant contributions in IoT security protocols, cloud-edge security for smart cities, and cyber-physical system resilience.

The impact of such authors is also evident in their citation impact, which demonstrates the high academic and practical significance within urban cybersecurity designs. As smart city cybersecurity challenges change over time, these scholars remain at the forefront of producing novel security approaches, threat mitigation measures, and AI-based defense technologies.

Additional network visualization analysis will investigate co-authorship patterns, institutional partnerships, and interdisciplinary research connections to detect new research clusters and future trends in cybersecurity for smart cities and intelligent transportation systems.

The h-index (453), shown in Fig. 4, quantifies the citation influence of the top fifteen authors which sought to address the research question RQ 2. The measure assesses both the productivity and citation influence of researchers, and thus it is an important

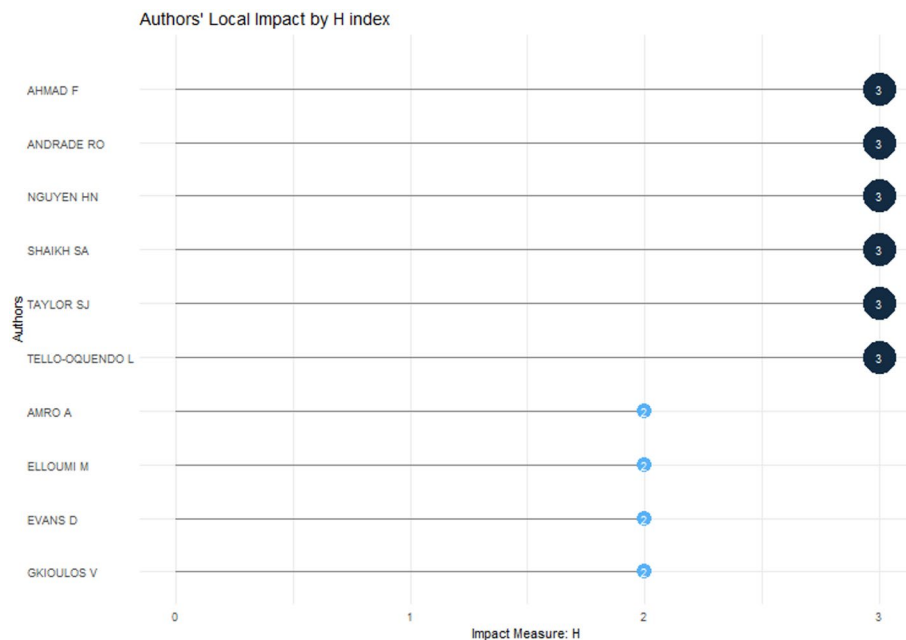


Fig. 4 H-Index for Top 15 Authors. Source: The authors (using Biblioshiny)

measure of scholarly impact. The h-index is defined as the maximum number of papers (h) that have received at least h citations each, so that both volume and impact of research contributions are considered.

An examination of the distribution of the h-index of cybersecurity experts in smart cities and ITS points to three degrees of authorial influence. High-profile researchers hold an h-index higher than ten, signaling intense and prolonged impact in research. Such authors have made pillar studies, whose major citations thereon appear over a series of studies. A middle level consists of authors possessing an h-index between 6 and 10, which portrays huge contributions that show moderate citations impact.

The citation trends with higher h-index values and publication counts highlight the influence of continued research efforts toward the formulation of cybersecurity policies and technological progress. The rising citation trends among high-index authors indicate an increasing academic dependence on their research outputs, which are instrumental in formulating security frameworks for smart cities and ITS.

Figure 5 provides details of the top authors' production in this domain. The trends in production reveal that while some authors have been producing work on a steady basis during the decade, others have appeared in recent years with high-impact publications.

The longitudinal productivity analysis of authors reveals that the initial years of research in smart city cybersecurity were characterized by the emergence of studies on IoT vulnerabilities and risk management frameworks. The years went by, and subsequently, the emphasis began to be laid upon AI-based security mechanisms, blockchain-based authentication, and real-time cyber threat detection in intelligent transportation systems. Pioneering authors in the field have evolved their research focus areas to include the above technologies to remain relevant in cybersecurity research.

Generally, Fig. 6 gives a straightforward trend of how contributor work has influenced cybersecurity research in ITS and smart cities, which sought to address the research question RQ 1, RQ 2 and RQ 3. The consistent efforts of prominent authors,

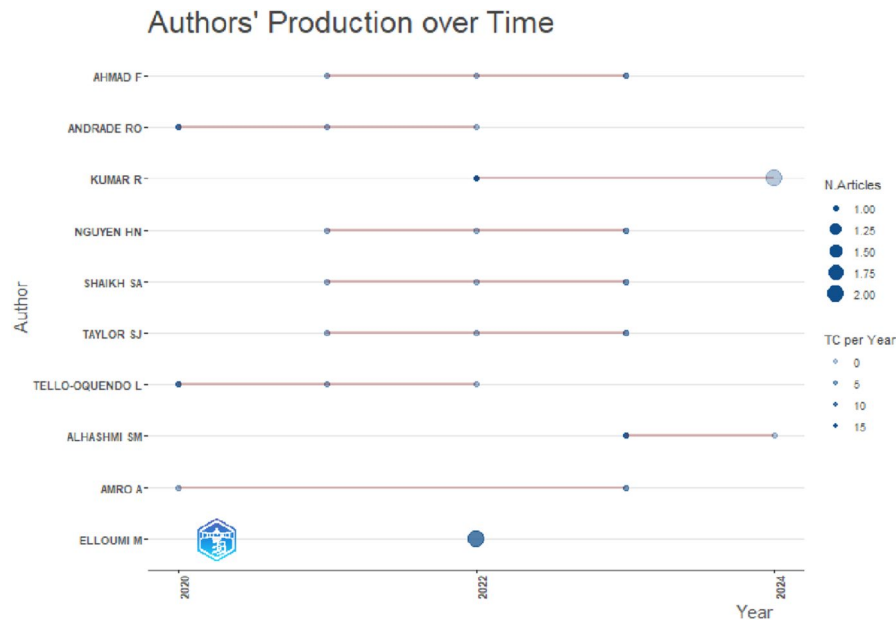


Fig. 5 Top authors production over time.

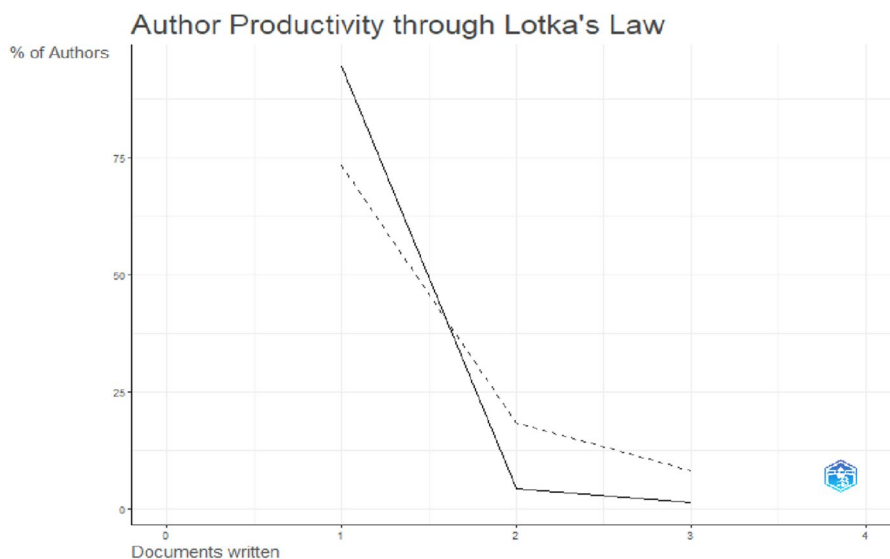


Fig. 6 The frequency distribution of authors productivity.

complemented by the entry of new experts, suggest a mature and thriving sector that keeps on increasing as it addresses new cyber threats and technologies.

The source growth trend analysis (Fig. 7) indicates a steady rise in publications across various journals, with IEEE Access, Lecture Notes in Networks and Systems, Lecture Notes in Computer Science, Smart Cities Cybersecurity and Privacy, and Sustainability (Switzerland) being the most prominent which sought to address the research question RQ 2 and RQ 3. Among them, IEEE Access has the highest cumulative occurrences, making it a key publishing platform for cybersecurity and AI-related research. Significantly, the initial dominance of IEEE Access between 2015 and 2019 was succeeded by a steady rise in multidisciplinary journals, especially in fields concerning sustainable

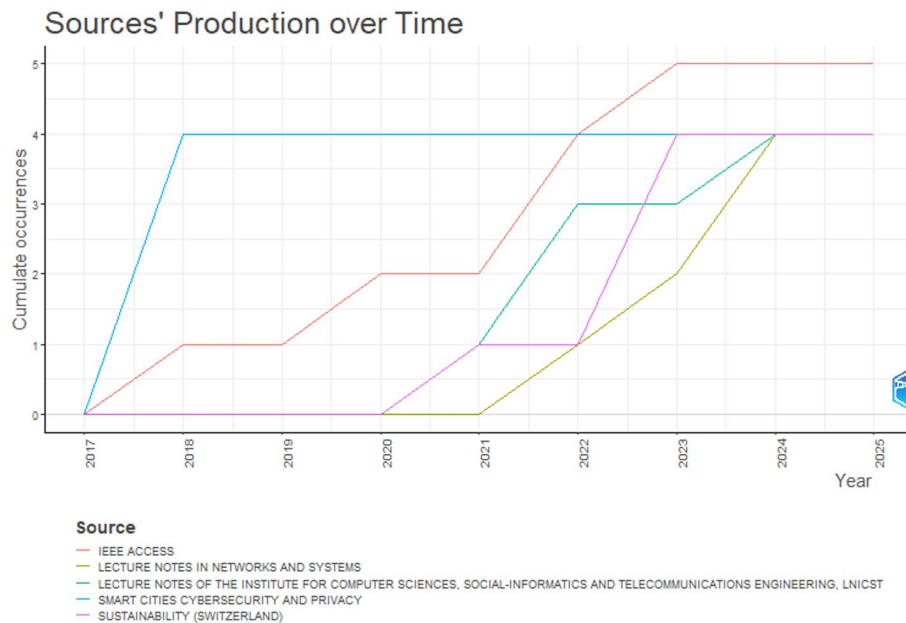


Fig. 7 Source growth trend across time. Source: The authors (using Biblioshiny)

digital transformation and AI-based security solutions. The dramatic rise in publications after 2020 corresponds with worldwide developments in AI-based cybersecurity solutions and regulatory policies, as there is growing emphasis on cyber resilience and automated threat intelligence processes (Scopus Dataset, 2015–2025). This trend correlates with previous studies indicating that the rapid adoption of AI in cybersecurity post-2020 is a response to the rise in cyberattacks and adversarial AI threats (Scopus Dataset, 2023).

The source growth trend analysis reveals a steady increase in the number of publications in IEEE Access (Fig. 6) which sought to address the research question RQ 2, Lecture Notes in Networks and Systems, Lecture Notes in Computer Science, Smart Cities Cybersecurity and Privacy, and Sustainability (Switzerland) from 2015 to 2025. IEEE Access exhibits the highest cumulative occurrences, indicating its role as a major publishing platform for cybersecurity and AI-related research. The early dominance of IEEE Access (2015–2019) is followed by a gradual rise in multidisciplinary sources like Sustainability (Switzerland) and Smart Cities Cybersecurity and Privacy, reflecting the integration of cybersecurity with sustainable digital transformation strategies.

A total of 615 publications seem to have been considered for review, with the understanding that 2025 data remain partial, as indexing for the year is still ongoing. The significant increase in publications post-2020 aligns with global advancements in AI-driven cybersecurity solutions and the growing regulatory emphasis on ethical AI governance (Scopus Dataset, 2015–2025). This correlates with existing literature, where the increased volume of publications post-2020 reflects industry-driven initiatives and academic research efforts addressing cyber resilience and adversarial AI threats.

Geospatial pattern of research contribution (Figs. 8 and 9) which sought to address the research question RQ 2 and RQ 4, depicts that the United States, India, and China are at the top of scientific output and making huge contributions towards cybersecurity research. The U.S. is predominant, supported by institutional support, federal research initiatives, and academia-industry collaboration. The increasing contribution from India

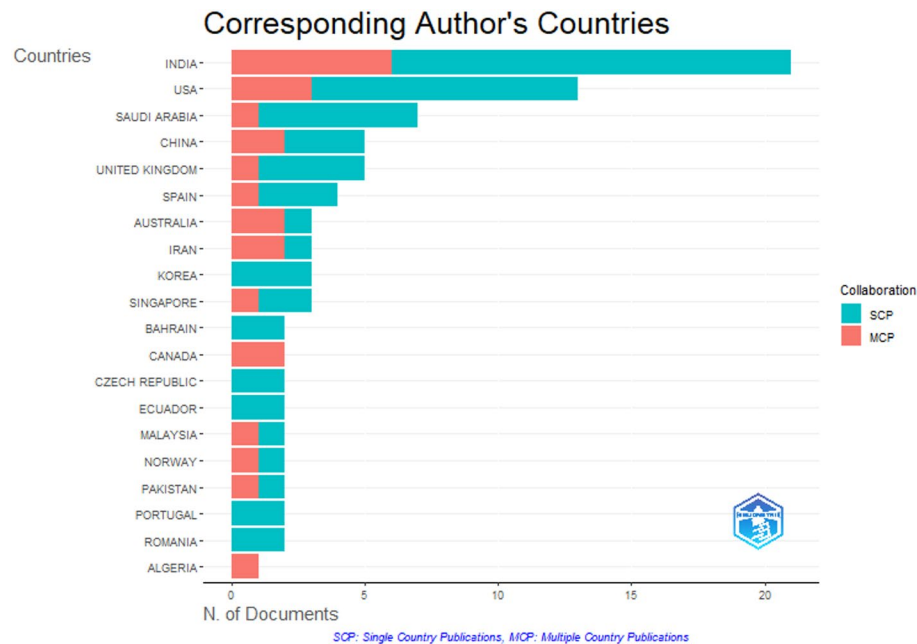


Fig. 8 Author country and multi-country author publications. Source: The authors (using Biblioshiny).

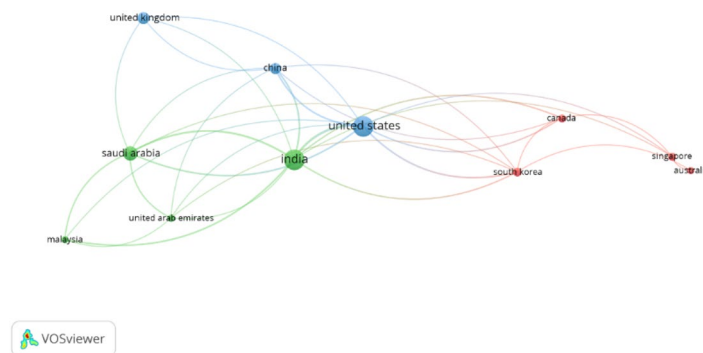


Fig. 9 Global country collaboration map. Source: the authors (using Vosviewer)

indicates its focus on AI-based security frameworks, regulatory policy, and training programs.

European countries like Germany, the UK, and France have moderate research activity, centered on GDPR compliance, privacy-enhancing technologies (PETs), and AI-based risk assessment models. Developing countries, especially those in Africa and South America, have limited submissions, as indicated by the digital divide and cybersecurity funding shortages (Scopus Dataset, 2015–2025).

These findings align with global cybersecurity reports that emphasize increasing collaboration between academia and industry to enhance cyber resilience strategies, particularly in high-risk domains such as critical infrastructure security and IoT-based cyber defense systems.

4.5 Network analysis and graphical visualization

The Global Country Collaboration Map produced by VOS viewer which sought to address the research question RQ 2 and RQ 4, illustrates co-authorship trends between

countries from a dataset of 170 papers on cybersecurity in smart cities and intelligent transport systems (see Fig. 9). The structure of the network indicates the level of global research collaboration across this set of articles. Given the limited dataset size, the observed network density might be less extensive than a broader analysis encompassing all publications in the field. Prominent nodes, representing countries with higher publication output within this set, should be considered in conjunction with earlier analyses of corresponding author countries to understand the distribution of research effort. Likewise, link thickness, representing the intensity of cooperation in terms of co-authored papers in the 170 articles, emphasizes major partnerships. Nevertheless, even highly visible links must be interpreted with care owing to the limited scope of the dataset. Cluster development indicates sets of countries working together intensively within this article sample, a function of regional location or mutual research interest in particular sub-domains of cybersecurity.

Geographical trends in collaboration, if realized, must be interpreted within the context of the 170-article dataset and not taken as representation of larger international trends.

4.6 Citation analysis

The citation counts for intelligent city, and ITS cybersecurity studies show a linear increase from 2017 to 2024 (see Fig. 10), with total citations of 3000 during the entire period which sought to address the research question RQ 2. The increasing trend of the citations is further indicative of the rising influence of the publications on the topic. In particular, the data reveal that the number of citations was lower than 1 in 2015, reaching approximately one hundred by 2018; it further increased to 200 in 2019, 300 in 2020, and around 450 in 2021. A more substantial jump was seen in 2022, with about 750 citations; this was followed by a tremendous increase in 2023 to about 1100 citations, which peaked at almost 1200 citations in 2024. Such an increasing trend of 3191 total citations signifies the growing impact of research in this area. While publication productivity was still low in 2015 and 2018, the citation numbers were still lower than that too; hence, their numbers conform to a typical lag in accumulating citations after a publication. Citation bursts from 2023 to 2025 reflect an increasing interest in AI-supported threat detection, zero-trust security, and the like, as new technologies are becoming

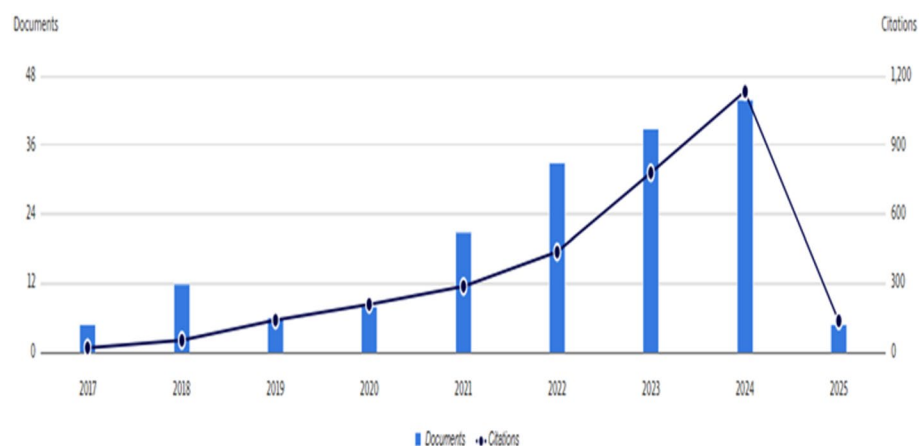


Fig. 10 Citation trend analysis (Source: The authors (using Scopus database))

more accepted in securing smart city systems. Drones security, protecting by edge-AI IoT devices, and privacy-preserving data analytics are also coming into the spotlight due to higher 5G network capacity and intelligent transport systems. These new topics are already pointing the way to a fundamental transition in modern urban security with the adoption of advanced, decentralized, and AI-based solutions. The sharpest rise in both publications and citations, between 2022 and 2024, indicates a possible increasing research activity and therefore impact. But caution should be exercised for 2025, with only a few citations appearing, due to the nature of those figures being incomplete since both publications and citations would, in all hoods, have low value when the research was still pending dissemination and citation. What correlation could be established between publication output and citation trends suggests a lag effect which sought to address the research question RQ 4, with publication peaks often-occasions preceding citation peaks, which occurs due to the time it takes for the research publications to be absorbed and cited.

The keyword analysis visualized as a tree map (see Fig. 11) shows the details of some important themes and concepts from the 170 articles explicitly focused on cybersecurity in smart cities and intelligent transport systems which sought to address the research question RQ 1. “Cybersecurity” and “smart city,” are represented with the greatest number of occurrences: eighty-one times each (9% against the entire keyword occurrence), to assert the point at the focus of the research. “Internet of things” appears next in prominence among the major keywords with fifty-five items (6%) to denote the essential role played by IoT devices and networks in smart city infrastructure, facing associated security challenges. “Network security” is mentioned fifty times (6%) to assure the importance of securing communication networks and data flows across these complex systems.

A number of keywords reference specific security concerns and methodologies. “Cyber-attacks” (20 times, 2%) and “intrusion detection” (16 times, 2%) reflect the interest in the identification of such nefarious activities and subsequent mitigation. “Risk assessment” (13 times, 1%) accentuates the need for risk management in dealing with cybersecurity challenges in the smart city context. “Cryptography” (14, times 2%) and “blockchain” (11 times, 1%) suggest high-tech security techniques in encryption and distributed ledgers in the protection of data and system integrity. “Data privacy” (14 times, 1%) and “privacy” (8 times, 1%) evoke the crucial protection of sensitive information in data-driven urban contexts.

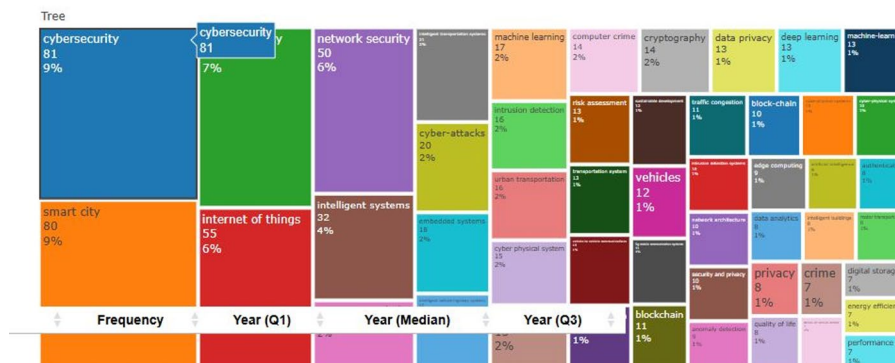


Fig. 11 Treemap of author keywords. Source: The authors (using Biblioshiny)

The tree map also reveals keywords related to specific smart city domains and technologies. “Intelligent systems” (32 occurrences, 4%) reflects the growing use of AI and other intelligent technologies in urban management and operations. “Embedded systems” (13 occurrences, 2%) emphasizes the security considerations for resource-constrained devices often found in smart city deployments. “Urban transportation” (13 occurrences, 1%) and “vehicles” (12 occurrences, 1%) point to the specific challenges of securing intelligent transport systems and connected vehicles. “Cyber-physical systems” (13 occurrences, 2%) acknowledges the interconnectedness of digital and physical infrastructure in smart cities and the need to secure these integrated systems.

The network visualization of keyword co-occurrence (see Fig. 12) reveals key thematic clusters within the 170 articles on cybersecurity in smart cities and ITS which sought to address the research question RQ 1. The output of the keyword co-occurrence analysis via VOS viewer presents three predominant thematic clusters that influence the research field of cybersecurity in smart cities and intelligent transportation systems. The first cluster elaborates AI-driven cybersecurity, where the terms artificial intelligence, machine learning, deep learning, and intrusion detection systems are accompanied by a strong emphasis on intelligent threat detection and the automated defense mechanisms. The second cluster indicates research on intelligent transportation and IoT-enabled mobility which is depicted by the keywords like intelligent systems, autonomous vehicles, urban transportation, and 5G communication systems, thus, pointing out the growing concern of securing cyber-physical mobility networks and connected transport infrastructures. The third cluster denotes cyber-physical and blockchain-based security with terms such as cyber-physical systems, embedded systems, blockchain, and network architecture indicating the decentralized security models and the hardware-level protection in smart city environments standing out. In general, the clusters mark a cross-disciplinary movement towards AI-powered security, resilient smart mobility solutions, and blockchain-enabled protection in urban digital ecosystems.

4.7 Content analysis

The keyword statistical analysis, network visualization, PageRank analysis, and thematic development which sought to address the research question RQ 1 and RQ 3 give a

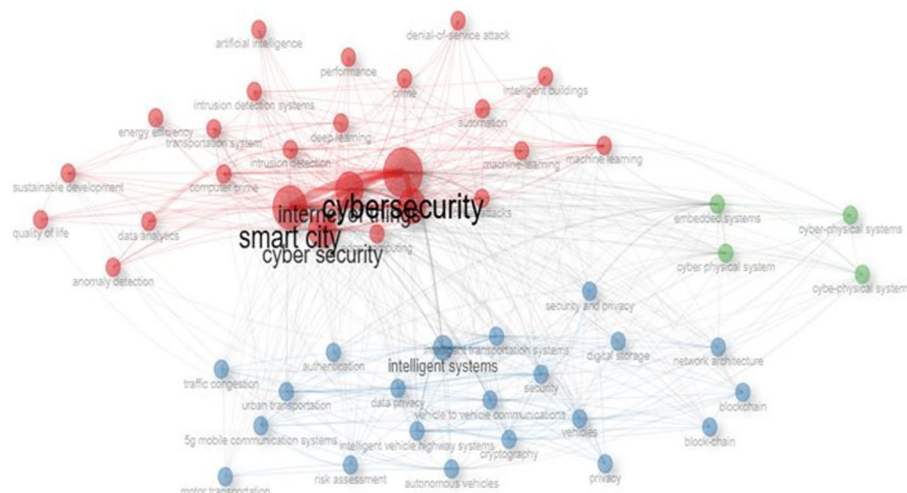


Fig. 12 Network visualization of keyword co-occurrence analysis (Source: The authors (using VOSviewer))

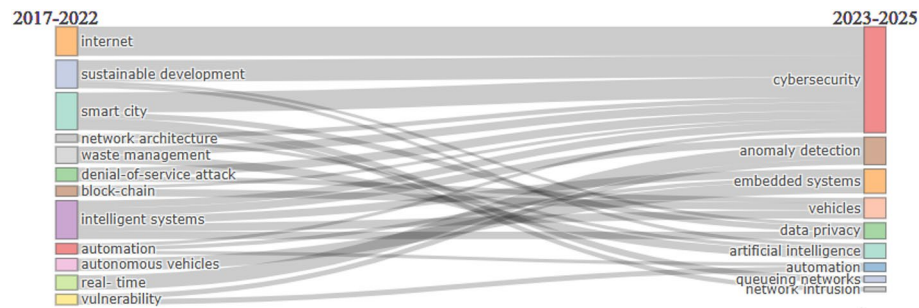


Fig. 13 Theme evolution. Source: The authors (using Biblioshiny)

Table 2 Page rank of author keywords. Source: The authors (using Biblioshiny)

Node	Cluster	Betweenness	Closeness	PageRank
cybersecurity	1	224.715	0.020	0.095
smart city	1	151.896	0.020	0.079
cyber security	1	101.186	0.020	0.063
internet of things	1	102.631	0.019	0.064
network security	1	79.439	0.019	0.063
cyber-attacks	1	4.095	0.014	0.023
intrusion detection	1	4.417	0.013	0.022
machine learning	1	3.804	0.014	0.020
computer crime	1	1.421	0.013	0.018
deep learning	1	0.669	0.012	0.011

comprehensive picture of the research environment across the 170 articles on smart city cybersecurity and ITS (see Fig. 13).

“Cybersecurity” and “smart city” are the dominant keywords, occurring most frequently, being most central, and exercising most influence as indicated by their high PageRank values (0.095 and 0.079, respectively) (see Table 2) large node sizes, and strong connectivity in the network visualization. This dominance is also compounded by their leading positions in the tree map, indicating their central position in the research conversation. “Internet of things” and “network security” also have high PageRank scores (0.064 and 0.063), as expected of their status as key themes, in accordance with their high frequency and leading positions in the treemap and network view. All these terms, and “cyber security” (PageRank 0.063), are part of a closely grouped cluster, embodying a thematic research interest in protecting interconnected devices and networks in smart cities. Being present is the word “internet of things”, highlighting the characteristic issue raised by the growth in connected devices, and “network security” for highlighting the requirement to defend communication infrastructure.

The PageRank scores, along with node size and network links, show the comparative influence of every keyword in the network, and how “cybersecurity” and “smart city” as hubs link key elements of the research. These are confirmed further by the results of cluster analysis of the network visualization, displaying specific research areas based on major cybersecurity issues, intelligent systems and data protection, and cyber-physical infrastructure. The thematic development, examined by time slices (2015–2022 and 2023–2025), illustrates how these themes have developed and changed. For example, earlier emphasis on more general smart city themes such as “internet,” “sustainable development,” and “network architecture” shifts to more precise cybersecurity issues during the later timeframe, with a greater concentration on “cybersecurity,” “anomaly

detection,” “embedded systems,” “data privacy,” and “artificial intelligence.” This transition demonstrates the increased maturity of the topic and greater specialization in resolving individual security issues. The advent of “anomaly detection,” “embedded systems,” and “data privacy” during the latter period is evidence of growing needs for real-time threat detection, protecting resource-limited devices, and safeguarding sensitive data. The increased visibility of “artificial intelligence” indicates expanding interest in harnessing AI to improve cybersecurity.

Through the integration of keyword statistics, network visualization, PageRank analysis, and thematic evolution, the overall picture of the research environment is revealed with the dominant themes, their relationship to each other, their relative significance, and their development over time in the 170 sampled articles. The high PageRank values of the core keywords, combined with their robust network affiliations and prominent locations in the treemap, reinforce their influence in determining the research agenda. The thematic development demonstrates an ever-changing field evolving to meet the new challenges and opportunities, and there is clearly a trend toward more specialized and advanced methods to cybersecurity in ITS and smart cities.

5 Discussion

The focus of this paper was to map the cybersecurity literature in smart cities and intelligent transport, in broad bibliometric perspective, into prominent research questions. To the knowledge of the authors, intellectual mapping of cybersecurity scholarly work within urban innovation has so far been under-researched. The results below represent major findings and responses to the research questions.

5.1 Restructuring of cybersecurity research trends and global contributions (RQ2)

5.1.1 Evolution of research and citations

The bibliometric analysis points to the remarkable evolution and increase in cybersecurity research on urban innovation. Addressing RQ2, the bibliometric data reveal a 150% surge in publications from 2021 to 2024. This exponential growth is causally linked to the post-pandemic acceleration of digital transformation, where cities were forced to adopt remote management technologies, thereby expanding the attack surface for cyber threats. The peak of 44 articles in 2024 suggests that the industry has moved from theoretical modelling to urgent implementation of security protocols. Notably, the trajectory of citations has grown, with the highest point in terms of smart city research related to cybersecurity occurring in 2022. In this timeframe, blockchain was heralded as a revolutionary technology that not only provided decentralization as a trust framework but also offered immutable records as a backing of data to enhance the cybersecurity of smart city systems. This is an indicator of increased awareness of the significance of the field, which runs in tandem with developments in IoT, AI, and blockchain technology use in cities [16].

5.1.2 Influential research contributions and authors

The influential research work mapping shows that relatively few researchers have authored a considerable number of high-impact papers. Fewer than 15% of authors have published more than three papers on cybersecurity in smart cities, consistent with the outcomes of previous bibliometric reports on emerging technology fields. Additionally,

the highly cited papers (with more than 500 citations) were mostly printed between 2015 and 2020, which indicates that seminal work in this field has found intellectual salience over time.

5.1.3 Geographical distribution of research

The geospatial analysis reveals distinct policy drivers behind research output. The dominance of the USA and China reflects a ‘technical-first’ emphasis, focusing on infrastructure scalability and surveillance, evidenced by the high frequency of ‘IoT’ and ‘Intelligent Systems’ keywords. In contrast, research from the European Union (e.g., the UK, Spain, Germany) shows a strong correlation with keywords such as ‘Data Privacy’ and ‘GDPR’ [17]. This indicates a causal relationship between the EU’s stringent regulatory environment (General Data Protection Regulation) and the academic emphasis on privacy-preserving architectures, which diverges from the deployment-centric focus of the Asia-Pacific region.

5.1.4 Institutional research influence

Most of the high-impact cybersecurity research is generated from technical universities and research organizations with specialization in cybersecurity and urban planning. Prominent players in this list include MIT, Tsinghua University, and research programs run by the European Commission. Publishing smart city cybersecurity research in conventional cybersecurity and engineering journals adds emphasis to its increased relevance.

5.2 Evolution of threats and emerging security frameworks (RQ 1 and RQ3)

With the swift adoption of digital technologies across urban spaces, cybersecurity research has branched into multiple methodologies, thematic fields, and research clusters:

5.2.1 Methodological evolution

Cybersecurity for smart cities includes a combination of technologies that have emerged to address the extraordinarily complex challenges of insecure devices and networks, which can lead to unbounded attacks. In this section, we highlight the issues discovered by research and their proposed solutions [18].

5.2.2 Core research themes and clusters

Regarding RQ1 (Threats) and RQ3 (Frameworks), the keyword evolution explains the shift in security strategies.

From Network Security to AI (RQ3): The rise of ‘Artificial Intelligence’ keywords is a direct response to the volume of data generated by 5G networks, which human analysts can no longer monitor manually. This has necessitated the shift toward AI-driven anomaly detection.

The Blockchain Shift (RQ3): The emergence of ‘Blockchain’ (11 occurrences) addresses the specific threat of centralized failure points. The data suggests a causal move toward decentralized ledgers to ensure data integrity in untrusted environments, such as V2X (Vehicle-to-Everything) communications.

Table 3 Summary of key thematic clusters in urban cybersecurity research

Serial No	Cluster	Examples
1	AI-driven cybersecurity	Anomaly Detection: Machine learning algorithms that detect irregular patterns in real-time traffic or data flow to identify attacks. Automated Defense: AI-driven response solutions that can counter threats without human intervention.
2	Intelligent transportation and IoT-enabled mobility	V2X Security: Securing Vehicle-to-Everything communications to prevent remote hijacking of autonomous vehicles . Connected Infrastructure: Protecting smart traffic lights and public surveillance networks from intrusion.
3	Blockchain-based security	Data Integrity: Using immutable records to back up data and prevent tampering in smart city systems . Identity Management: Blockchain-based authentication protocols to verify users and devices.

The thematic progression of cybersecurity research in smart cities can be grouped into the following major clusters: Table 3. Summary of Key Thematic Clusters in Urban Cybersecurity Research.

5.2.2.1 Artificial intelligence and cybersecurity Cybersecurity solutions developed using artificial intelligence, such as machine learning-driven anomaly detection, are becoming imperative countermeasures against cyber threats in intelligent transportation networks [19].

5.2.2.2 Blockchain in case of secure smart cities Security models based on blockchain have been investigated for securing sensitive data of various cities and maintaining trust and dignity in decentralized smart city systems [20]. Smart cities have appeared in Japan, the UAE and Germany, where these cities constitute an excellent future environment in which they can live and are more suitable than ordinary cities. In this report, the most critical challenges that cybersecurity faces in preserving smart cities from hacking operations will be reviewed in general [21].

5.2.2.3 Privacy and ethical issues Ethical issues, data privacy, and surveillance implications in smart cities have been areas of growing research [22].

5.2.3 Cybersecurity models and decision-making frameworks

The cybersecurity models used in smart cities can be divided into three main models:

Risk-Based Models: Allocating cybersecurity investments in accordance with the risk analysis of urban infrastructure [23].

Multi-Layered Security Strategies: Integrating network security, AI-powered monitoring, and blockchain-based identity management for comprehensive protection [24].

Cyber Resilience Strategies: Investigating urban cybersecurity readiness and response processes in the case of cyberattacks [25].

5.2.4 Impact of external and internal factors on cybersecurity decision-making

The research determines major external and internal factors influencing cybersecurity in smart cities and intelligent transport:

External Factors: Cyber-attacks, regulatory regulations, geopolitical uncertainty, and technological developments have significant effects on urban cybersecurity systems.

For example, global cybersecurity policies like the EU's Cybersecurity Act have affected national security strategies [26].

Internal Factors: Organizational cyber culture, investment in cyber infrastructure, and stakeholders' participation in urban innovation make for successful cyber resilience.

5.3 Critical gaps and challenges in urban digital infrastructure (RQ 4)

The bibliometric analysis also picked out main gaps and upcoming research topics that should be addressed by future research:

5.3.1 Integration of quantum cryptography and blockchain in smart cities

Quantum cryptograph methods are receiving interest as a remedy to counter forthcoming cybersecurity threats within urban networks. Future research needs to investigate the applicability of quantum key distribution (QKD) in smart city infrastructures [27]. Future research can also be done to investigate use of blockchain in smart city infrastructure sustenance.

5.3.2 Cybersecurity regulations and policy implications

There is a pressing need to analyze the efficacy of cybersecurity policies in smart cities, especially in the aspect of compliance with international cybersecurity standards. Comparative studies regarding cybersecurity frameworks across different regions would be valuable. Future research should examine how differing regulatory environments (e.g., EU vs. Asia-Pacific) impact the technical implementation of security protocols [28].

5.3.3 Human-centered cybersecurity approaches

Whereas most cybersecurity studies aim at technicality, there is a requirement to investigate the human element, such as user conduct, consciousness, and the involvement of social engineering in smart city cyber threats [29, 30].

5.3.4 Cybersecurity for autonomous transport systems

The speedy introduction of autonomous cars (AVs) in cities is of cybersecurity concern. Research in the future should concentrate on Vehicle-to-Everything (V2X) communication network security and how cyber threats will impact autonomous mobility [31].

6 Conclusion and implications

As concluded from the bibliometric evaluation, the sphere of cybersecurity research in smart cities and intelligent transport seems to be growing exponentially. The research highlights the growing academic focus on security issues in urban innovations involving IoT, AI systems, and blockchain systems. The heightened use of digital infrastructure in smart cities calls for stringent cybersecurity practices to safeguard against cybercrime and guarantee operational resilience. The research suggests substantial prospects for future research, especially in regulatory frameworks, risk mitigation mechanisms, and security solutions with AI and blockchain.

Furthermore, the forecast is that the fusion of blockchain technology will be able to bring about the threefold benefits of trust, transparency, and decentralized security in the administration of data transactions in smart city ecosystems.

Theoretically, this research identifies major research themes and their trajectory and emphasizes IoT security, AI-based cyber defence, blockchain for secure payment, and smart governance models. The bibliometric analysis establishes the key influencers, institutions, and citation patterns that have influenced this field. Policy Implications: Based on the identified gaps (RQ4), three specific policy actions are recommended: Harmonisation of Standards: The divergence between EU privacy-focused research and US/China technical research creates security gaps. Policymakers must establish a unified 'Smart City Security Standard' that merges GDPR-style privacy with technical scalability.

Mandated Algorithmic Audits: Given the reliance on AI, city administrators must enforce mandatory audits of AI algorithms to prevent adversarial attacks on transport infrastructure.

Legacy Infrastructure Funding: Policies must specifically allocate budget for retrofitting legacy systems, which remain the most vulnerable entry points for cyberattacks.

7 Limitations of the study and future research areas

One of the limitations of the study arises from the fact that only SCOPUS database was used for extraction of articles hence any articles from other databases have been missed out. However future studies can be done using other databases like Web of Science, Science Direct etc. Based on the study the following areas can be undertaken for future studies:

7.1 Quantum-resistant cryptographic solutions

Future research could be done to investigate the applicability of quantum key distribution (QKD) in smart city infrastructures. Future research can also be done to investigate use of blockchain in smart city infrastructure sustenance.

7.2 AI-driven predictive security models

There is a requirement to investigate AI-driven predictive security models; the involvement of social engineering in smart city cyber threats. Research in the future should concentrate on Vehicle-to-Everything (V2X) communication network security and how cyber threats will impact autonomous mobility.

7.3 Cross-sector regulatory initiatives

Comparative studies regarding cybersecurity frameworks across different regions would be valuable. Future research should examine how differing regulatory environments (e.g., EU vs. Asia-Pacific) impact the technical implementation of security protocols.

Author contributions

1. Sujata Joshi: Ideation, design of the work; drafting, editing, revising, supervision. 2. Akshay Baviskar: Drafting, acquisition, analysis, and interpretation of data, revising manuscript. 3. Sarvesh Rajmane: Drafting, acquisition, analysis, and interpretation of data, revising manuscript.

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Data availability

All data generated or analyzed during this study are included in this published article (and its Supplementary Information files). The data sets generated and analyzed during the current study are also available from the corresponding author on reasonable request.

Declarations

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Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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